
Supporting information

AI as an Amplifier in Chemistry Laboratories: A SERG-Based Framework for Embodied and Assessable Instruction

Zhaohong Zuo^{a,*}, Jun Li^a

5 ^a School of Chemistry and Chemical Engineering, Chongqing University, Chongqing 401331, China
(ORCID 0000-0002-2460-8702)

* E-mail: zzhong@cqu.edu.cn

Appendix S6 (SI). Compliance Documents (Consent, Privacy & Data Management)

- 10 **Scope.** For use in the project “AI-Enhanced Teaching and Implementation in Chemistry Laboratory Classes.” This packet is **not legal advice**; adapt to your institution’s policies and applicable laws (e.g., **FERPA**, **GDPR**) before use. All square brackets **【】** are placeholders.

S6.0. Usage Notes & Disclaimer

- 15
- **Coverage:** Participant information & consent, media/work release, **Privacy & Data Management Plan (PDMP)**, **Record of Processing Activities (ROPA)**, de-identification & public-release rules, safety & incident response card.
 - **Principles:** Data minimization; purpose limitation; encrypted storage; auditable access; scheduled deletion; prior consent; de-identified public release.
 - **Governance:** Course Lead (**PI**) holds overall responsibility; IT/Legal/Ethics offices participate in approvals.

20

S6.1. Participant Information & Consent (adult/minor versions)

S6.1.1 Participant Information Sheet

- 25
- **Project title:** **【AI-Driven Improvement and Evidence-Based Research in Chemistry Laboratory Teaching】**
 - **Lead (PI) / Contact:** **【Name / Dept. / Email / Phone】**
 - **Purpose:** Improve stability and learning quality in lab classes; evaluate the effectiveness of integrating AI and pedagogy.
 - **What participation involves:** Regular coursework, data and work submission, optional interviews/recordings; class imagery may occur.
 - **Potential risks:** Network/device interruptions; all materials de-identified before public release.
 - **Potential benefits:** More stable support, clearer feedback, and curated learning resources.
 - **Data items (minimum necessary):**
 - **Basic:** Class, student ID or random code (no names/IDs in public materials).
 - **Learning data:** Lab datasets, figures, reports, quiz scores, rubric ratings.
 - **Process evidence:** Class recordings/photos (faces may be blurred), logs, interview audio (optional).
 - **Use & retention:** For course improvement and academic research only. Raw data kept until end of term + 1 year; then de-identified for up to **5 years** or destroyed per policy.
 - **Voluntariness & withdrawal:** You may withdraw at any time; upon withdrawal, no new data will be collected and identifiable information will be deleted (except where required by law/archives).
 - **Privacy & security:** Encrypted storage, role-based access, audit logs; de-identification and compliance review before any public release.
- 40

S6.1.2 Consent Form — Adults (Students/Teachers)

I have read and understood the Participant Information Sheet. I voluntarily join this study/teaching improvement and agree to the following uses of my data:

- 45
- Course improvement and evaluation (**required**)
 - Anonymized/de-identified scholarly publications and conference presentations (**optional**)
 - De-identified class imagery/works for internal open classes and training (**optional**)

Signature: _____ Name **【】** Class/Dept. **【】** Date **【】**

S6.1.3 Parental/Guardian Consent for Minors (if applicable)

I am the guardian of **【Name】**. I have read the information and consent to participation and the selected uses:

- 50
- Course improvement and evaluation (**required**)
 - De-identified academic/internal display (**optional**)

Guardian signature: _____ Contact **【】** Date **【】**

S6.2. Media & Work Public-Release Authorization (separate license)

- 55 For additional permission to use materials in open classes/promotions/resource repositories; **de-identification is the default** (blurring/renaming/metadata removal).

I consent to the de-identified use of my imagery/works in:

- 60
- Institutional website / open-class repository
 - Academic conferences or article appendices (de-identified)
 - Promotional materials (**requires additional institutional approval**)

I understand I may withdraw future use authorization at any time; already released materials are handled per platform policies.

Signature: _____ Date **【】**

65

S6.3. Privacy & Data Management Plan (PDMP)

S6.3.1 Roles & Responsibilities

Role	Name/Dept.	Responsibilities
Course Lead (PI)	【】	Overall compliance & data responsibility; approvals & publication gatekeeping
Data Steward	【】	Data classification, storage, backups, audit logs
Compliance/Ethics Contact	【】	Consent review, privacy/safety training, incident response
Scoring & Research Team	【】	De-identification, dual-rating and κ logging

S6.3.2 Data Classification & Minimization

- 70
- **Level A (Restricted):** Identifiers (name/ID/raw face images/contact) → Local encrypted drives & controlled institutional vaults only.
 - **Level B (Internal):** Learning data & scores; interview transcripts → Authorized team access only.
 - **Level C (Public):** De-identified charts/posters/summary tables; no personal identifiability.

S6.3.3 Lawful Bases (examples)

- **Teaching improvement:** Legitimate educational interest / instructional duty.
- **Research outputs:** Informed consent (withdrawable; published materials handled per academic norms).

-
- 75
- Compliance with institutional rules and applicable laws (e.g., **FERPA, GDPR**).

S6.3.4 Data Flow & Storage (“text flowchart”)

Collection (class/platform) → **Local secure folder (encrypted)** → **De-identification** → **Tiered repository (A/B/C)** → Analysis & scoring (audit-logged) → Output only aggregate/anonymized results → **Second de-identification review** for public materials → Release & archive.

80

S6.3.5 Access Control & Logging

- Named-user, least-privilege access.
- All create/read/update/delete actions produce audit logs (time, account, purpose).
- External sharing requires an approval ticket and time-bound recall.

S6.3.6 Retention & Deletion (suggested; adapt to policy)

Data Type	Retention	Disposition
Raw identifiable imagery	End of term + 1 year	Delete or cold-store (approval required)
De-identified learning data	5 years	Archive or delete after review
De-identified interview transcripts	5 years	Archive or delete after review
Scoring & κ records	3 years	Keep summaries; delete raw
Approvals & consent forms	Per institutional archives rules	Dual (paper/digital) storage

85

S6.3.7 Third Parties & Cross-Border (if using cloud)

- List: **【LMS / Cloud drive / Analytics platform / Video platform】**; confirm data location, encryption, sub-processors.
- Cross-border transfers (if any) require separate assessment/approval and lawful mechanisms (e.g., SCCs/institutional agreements).

90

S6.3.8 Incident Response (summary)

- **Team:** PI + Data Steward + Compliance Contact + IT.
 - **Timelines:** Initial containment within **24h**; notification & remediation plan within **72h** (per policy).
 - **Record:** Incident type, impact scope, response steps, recurrence prevention.
-

95

S6.4. Record of Processing Activities (ROPA, short form)

Field	Content
Purpose of processing	Teaching improvement, course evaluation, academic research
Data subjects	First-year students, course instructors, peer observers
Data categories	Scores & works, lab data, interview transcripts, (optional) imagery
Lawful bases	Educational interest/duty; informed consent
Recipients	Course team; public: anonymized aggregates only
Cross-border transfers	【None / Yes: platform & safeguards】
Retention	See PDMP §3.6
Security measures	Encryption, tiered access, logs, de-identification, audits
Risk assessment	Medium; main risks: inadvertent disclosure / unauthorized access
Mitigations	De-identification, approvals workflow, training & drills

S6.5. De-Identification & Public-Release Standards (with checklist)

S6.5.1 Core Rules

- **Remove:** Names, student IDs, phone/email, precise locations, facial features.
- **Replace:** Student ID → random code (e.g., **S-0023**); Class → abstract label (e.g., **Class-A/B**).
- **Blur:** Dates to week/month; show scores as bands/percentiles.
- **Imagery:** Blur faces and identifiable backgrounds; remove EXIF/metadata.
- **Filenames:** No names/IDs; e.g., **CQU_AIchem_Year_S-XXXX_ReportV1.pdf**.

S6.5.2 De-Identification Checklist (pre-release self-check)

Item	Pass	Notes
No names/IDs/contacts in text/tables	☐	
Faces/IDs in images/videos masked	☐	
Metadata (EXIF/author) removed	☐	
Code/logs contain no personal data	☐	
Public files include “purpose & copyright” notice	☐	

S6.6. Safety & Incident Response Card (RAMP × InfoSec)

Lab safety (RAMP):

- **Recognize:** List chemicals/equipment and hazard classes.
- **Assess:** Concentration/dose/exposure routes & populations.
- **Minimize:** Ventilation/PPE/substitution; dual verification for high-risk steps.
- **Prepare:** Eyewash/extinguisher/spill response & contact info.

Information security:

- **Backups:** 3-2-1 strategy (3 copies, 2 media, 1 offsite).
- **Encryption:** Full-disk encryption locally; TLS in transit.
- **Passwords:** Strong passwords + MFA.
- **Least privilege:** Role-based; revoke upon team exit.
- **Vulnerabilities/incidents:** Contain within **24h**, notify within **72h** (per policy).

S6.7. Versioning & Approvals

- **Document version:** v **【 】** ; **Author:** **【 】** ; **Date:** **【 】**
- **Approvals:** Course Lead **【 Signature 】** ; IT **【 Signature 】** ; Legal/Ethics **【 Signature 】**
- **Effective date:** **【 】** ; **Review due:** **【 】**
- **Scope:** This course / This school / Institution-wide (circle one)

125 **“One-Click” Usage Suggestions**

- Export this appendix as fillable **PDF/Word**.
- Collect **S6.1/ S6.2** via online forms with e-signatures; auto-archive.
- Publish **S6.3–S6.6** to the project repository with **read-only** permissions.
- Before any public release, run the **S6.5 checklist** once more.

130